Support-conditioned screening of incomplete river-temperature years: retrospective evidence from 146 networks

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Key points

- Whether a gap matters depends on how much of the target statistic's support it hides; we use that as a screening axis, not a universal law
- Sealed USGS and multi-agency panels, each scored once, cut pooled designed-outage errors by 7.9 and 4.6 points at 70% retention
- On 2422 real incomplete years, peak-window and length rules disagree; wholly-hidden peak counts use a climatological proxy

Abstract

Thermal archives are usually screened by completeness thresholds that measure outage size, not the support of the target statistic. We develop a framework that conditions on how much of that support a gap hides, links it to reconstruction error and archive admission, and calibrates risk. On 5114 complete river station-years from 146 networks (51524 designed outages), a frozen hottest-30 screen on a retrospective 36-network holdout cuts consequential errors from 9.7% to 2.5% at 70% retention (about $[-8.97, -5.36]$ points); the empirical arm is unresolved. A sealed USGS panel outside the development programme (39 HUC4 networks; 8973 cases), scored once, recovers about -7.9 points ($[-10.2, -5.8]$), and about -9.4 points on the 15 networks with no outcome-bearing prior use; the empirical-arm gain is smaller but non-zero. On a pre-registered multi-agency transport panel — 32 networks from three agencies absent from the development programme, across boreal, oceanic and continental rivers, sealed and scored once — the same frozen screen gives -4.6 points ($[-6.5, -3.3]$), meeting the pre-declared rule; continuum, crossover and dilation results from the opened reserved panel stay exploratory outside both seals. On 2422 real incomplete years, a peak-window rule disagrees with length guidance on 15.1% of admissions and flags wholly masked *climatological* peak windows, a proxy whose positive predictive value for a wholly unobserved realised peak is 0.74 on 8973 sealed designed cases. In the United States subset, difficulty-based screening is associated with thinning high-elevation catchments; that association is weak and must be checked per archive rather than assumed.

Plain Language Summary

Rivers are getting warmer, and managers ask how hot a river got last summer. The answer needs a year of daily temperatures, but records have holes. Screening today mostly counts missing days. We argue the better question is whether the missing days are ones the number you want depends on, and we test a screening rule built that way on many river networks. On held-out networks with designed gaps the rule reduces large errors for the summer peak at equal retention; under the gaps these networks actually experience the gain is not yet clear. Results about exact crossover days, autumn maintenance benefits, and "peak never observed" counts are exploratory or use proxies and need confirmation before they guide operations.

1 Introduction

The quantity this paper is about is the error of a derived thermal statistic, not the error of a daily temperature series. Cold-water habitat assessment, thermal-regime classification and criterion-exceedance accounting all reduce a year of daily water temperature to a single number: a hottest 30-day mean, an annual mean, a seasonal amplitude (Caissie, 2006; Isaak et al., 2017). That number is what enters a management judgement, and a station-year is admitted to an archive or excluded from it on the strength of it. An error of one degree in a summer peak converts a thermally stressful year into an unremarkable one; on our panel that error is not hypothetical, because computing the peak from recorded days alone carries a mean absolute error of 1.305 °C, larger than one year-to-year standard deviation of the quantity itself.

The error of a derived statistic is not the same object as the error of the series it is computed from, and the difference is what makes the admission decision hard. A reconstruction with excellent daily skill can be badly wrong about a 30-day maximum if its errors are correlated inside the window that achieves the maximum, and a reconstruction with mediocre daily skill can recover that maximum exactly if the window survived. Daily skill is what the modelling literature reports; the derived statistic is what a manager reads. Between the two sits a decision that is rarely written down as an estimand-specific rule — whether this station-year may be used at all — which is made for every record in every thermal archive and is almost never made with the derived statistic in mind.

Two families of rule decide admission today, and they differ in what they assume rather than in when they were written (Table 1). Agency collection standards specify what must be present: Mauger et al. (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima". Quantitative guidance instead constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy" and that "imputation of missing data expanded this range to approximately 20 weeks". The gap-filling literature optimises the accuracy of the filled series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017), and the closest prior work

fills incomplete summer records so that a thermal peak can be computed, applies that filling uniformly, declines no station-year, and admits July–August-only records on the climatological position of the hottest window (Beaufort et al., 2022); it also quantifies the resulting bias of the thermal peak itself when whole years are removed, which is the closest precedent for the designed-outage measurement here. Irregular sampling and sampling artifacts already bias mean and trend estimates in stream-temperature series and identify which model structures stay trustworthy for means and trends under those artefacts (Grey et al., 2024), and long-term reconstruction frameworks have been developed for regulated and data-sparse basins including Chinese catchments and the Yangtze mainstem (Huang et al., 2023; Wang, 2025). Those studies answer prediction and reconstruction questions; they do not close the archive-admission question for a named annual extreme. Nor is statistic-awareness absent from practice: completeness conventions in climate monitoring are already defined per statistic, though by counts or seasons rather than by support (Zhang et al., 2011). What none of the thresholds in common use quantifies is how much of the target statistic's own support a record contains, and the transfer does happen in practice: Mauger et al. (2017), for example, retain a site-year for a mean July temperature only "when at least 90% of the days were captured", and then report the maximum weekly maximum temperature from the same records.

That is the assumption we examine, and its consequence is not confined to a prediction. An annual mean is a sum over days: removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme supported on a short window: if the outage misses that window the statistic is recoverable almost exactly however long the outage is, and if the outage covers it the statistic must be inferred however short the outage is. A single completeness threshold therefore cannot be correct for both, and the direction of its error is opposite in the two cases. Applied to a summer extreme, a threshold on gap length can admit station-years whose climatological peak window is wholly masked and discard station-years whose realised peak is intact; which of the two dominates is an empirical question about the archive, not a matter of taste. The consequence is an archive whose composition, and not only whose accuracy, depends on a choice nobody records.

Three things make the question answerable now that the studies that raised it did not combine. Water-temperature archives from nine national agencies can be assembled into a panel large enough to hold out whole river networks, so a screening rule can be

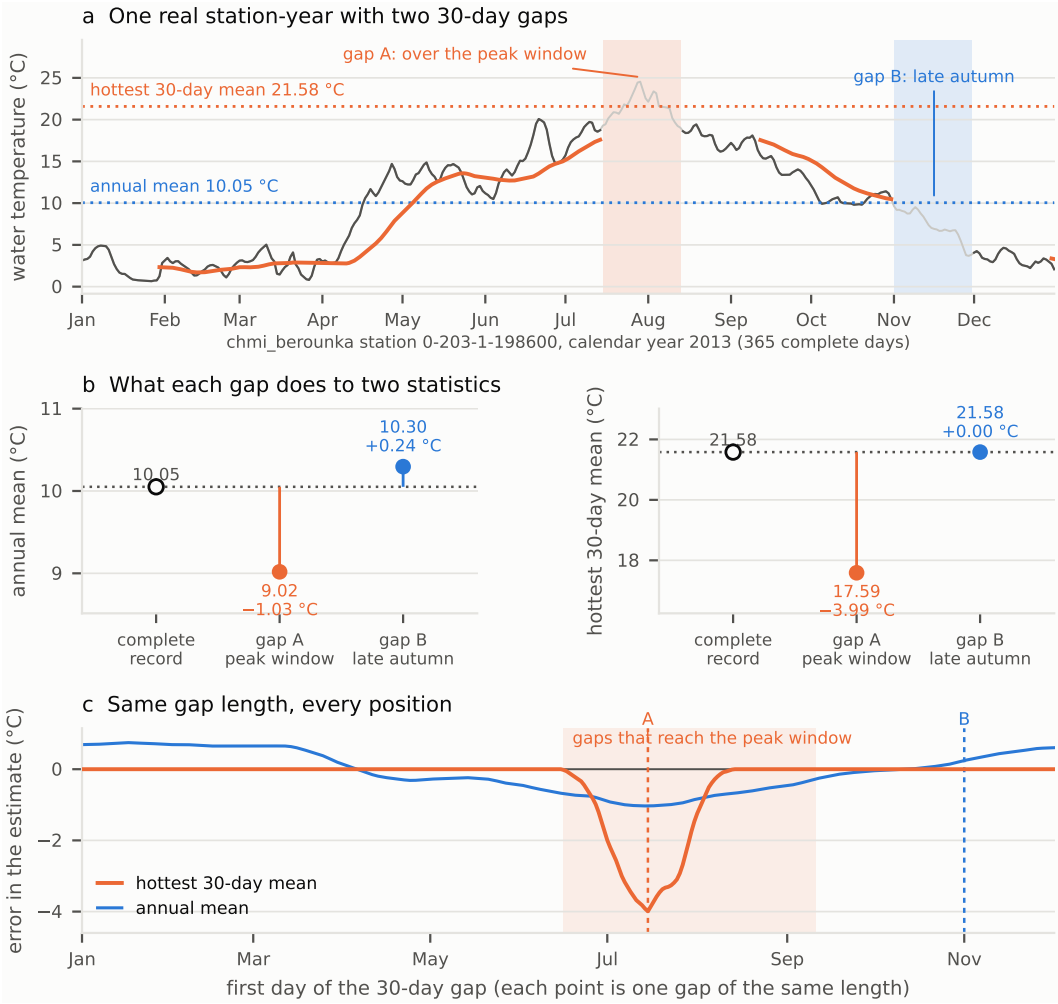


Figure 1. Two statistics, one record, two equal-length gaps. (a) A complete annual water-temperature record (CHMI Berounka, station 0-203-1-198600, 2013; n = 365 days, no missing day) with two designed 30-day gaps in different positions: gap A over the window that achieves the complete-record hottest 30-day mean (15 July to 13 August), gap B in late autumn (1 to 30 November). The curve is faint where the analyst would not see it and dark where the gap leaves it observed; the 30-day running mean is recomputed with both bands withheld, so it breaks wherever no fully observed window remains. (b) Each gap's effect on each statistic, in °C: dots are estimates from the days the gap leaves, open dots and dotted lines the complete-record truth (annual mean 10.05 °C, hottest 30-day mean 21.58 °C). The two panels use different vertical scales and neither starts at zero, so the printed values carry the comparison: gap A moves the hottest 30-day mean by −3.99 °C while the same length in late autumn leaves it exact, and the annual mean moves by −1.03 and +0.24 °C respectively. (c) Every one of the 336 possible 30-day gap starts of the same year, plotting the error each position induces in the hottest 30-day mean (orange) and the annual mean (blue); the shaded band holds starts whose window reaches the peak window. Position, not length, decides whether the peak survives. This is an illustration of the mechanism on one real record, not a result: no model, no population estimate, no test. The station-year belongs to the retrospective holdout split and enters no fit reported here.

scored on held-out river networks, and — after external recruitment — on networks outside the development programme. Because a fully observed calendar year supplies a truth, a gap can be placed on it deliberately, which converts the comparison between conditioning variables from an observational contrast into a designed one: the two candidate properties of a gap can be varied independently, which they never are in a record that failed on its own. And the share of a statistic's support that a gap hides is computable at decision time from the station's own earlier years, so the alternative conditioning variable is not merely definable but deployable by an agency with no data it does not already hold.

We therefore ask three linked questions, and the last matters most, because a screening rule is not evaluated by its discrimination but by the archive it leaves behind. First, which property of a data gap governs the error of a derived thermal statistic, and does the answer depend on which statistic is being computed? Second, if it does, is that dependence a dichotomy between two regimes or a continuous transition, and where does it turn over? Third, what does conditioning on the answer change — for the error among retained station-years, for which stations survive in the archive, and for the 2422 records in this same archive that really do have gaps and have never been decided?

2 Study system, data, and the quantity to be explained

2.1 The statistic, its support, and the share a gap hides

Let a station-year be a sequence $y = (y_1, \ldots, y_T)$ of daily mean water temperatures, $T \in \{365, 366\}$, and let $q(\cdot)$ be the derived statistic of interest. The family we sweep is the hottest consecutive k-day mean

$$q_k(y) \;=\; \max_{t} \; \frac{1}{k} \sum_{i=t}^{t+k-1} y_i, \tag{1}$$

which is the annual maximum at $k = 1$ and the annual mean at $k = T$; the coldest 30-day mean, the annual amplitude, and the sine-fit mean, phase and amplitude of Johnson et al. (2021) are scored alongside.

A gap is a binary mask $m \in \{0, 1\}^T$; the analyst observes the days where $m_t = 0$. A reconstruction $\hat{y} = R(y_{\bar{m}}, Z)$ completes the year using side information Z — synchronous donor stations, air temperature, the station's own history — and reports $q(\hat{y})$.

Table 1. What each body of work optimises, decides on, and conditions on. The rows are ordered by what they assume about the conditioning variable, not by date.

body of work	optimises	decision unit	conditioning axis
completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size (timing tested; effects reported inconsistent)
collection standards (Mauger et al., 2015)	comparability across a programme	deployment	season and duration
gap filling (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors
peak reconstruction (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air regression applied uniformly; climatological summer window used to admit partial-season records
large-sample models (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023)	prediction skill	day or reach	meteorology, catchment attributes
selective prediction (Geifman & El-Yaniv, 2017; Zaoui et al., 2020)	error at a fixed rejection rate	case	model confidence
this work	**error of the derived statistic among retained years**	**station-year**	**share of the statistic's support hidden**

The quantity that matters is the error of the reported statistic,

$$L \;=\; \big|q(\hat{y}) - q(y)\big|, \tag{2}$$

and a station-year is called *consequential* when $L > \tau$ for a tolerance τ in the units of q.

The property of the gap we propose as the governing one follows from how much each day moves the statistic. Writing $w_t^q = |\partial q(y)/\partial y_t|$ for the discrete influence function of q at y (Hampel, 1974), the *hidden mass* of a gap is the share of that influence it removes,

$$h_q(m) \;=\; \frac{\sum_{t:\, m_t=1} w_t^q}{\sum_t w_t^q}. \tag{3}$$

For the annual mean every w_t is equal and h_q is the missing fraction; for the hottest k-day mean the influence is concentrated on the k days achieving the maximum and h_q is the share of that window the gap hides; for the sine-fit parameters the weights are smooth functions of day of year — a constant for the fitted mean, $|\sin(\theta_t - \phi)|$ for the amplitude, which is largest at both extremes, and $|\cos(\theta_t - \phi)|$ for the phase, which is largest a quarter-year from the peak. Equation (3) therefore contains the incumbent axis as a special case rather than opposing it, which is why the two regimes reported below are ends of one scale.

Equation (3) cannot be evaluated at decision time, because the weights depend on the days the gap hides. What an analyst has is the *climatological* support: the window the station's other complete years say the maximum should fall in. That estimate has an error, and Section 4.5 shows the error is large enough to change what a screening rule should condition on.

2.2 Records, and what each dataset is for

Daily water-temperature series were assembled from nine national providers: ARSO (Slovenia), CHMI (Czechia), FOEN (Switzerland), GKD Bayern and LUBW (Germany), NVE (Norway), Rijkswaterstaat (Netherlands), SYKE (Finland) and the USGS (United States). Acquisition contracts differ, and we audited each one against the provider's own documentation (`results/wrr_submission_ready/measurement_provenance_dictionary.csv`). Five providers document a daily mean and the parser reads it: USGS (`parameterCd=00010`, `statCd=00003`), FOEN (`data_1day_mean`), GKD Bayern (`Tageswerte`, column `Mittelwert`),

NVE HydAPI (resolution 1440, documented as mean) and, for the transport panel below, the UK Environment Agency (measure period 86400, "daily mean Temperature"), with LUBW's `Tagesmittelwert` documented but its opaque workbook column only partially re-verified. Four providers do not meet that contract as read: CHMI's parser averages 10-minute *instantaneous* values (T_O) even though the provider also publishes a daily-mean product (T_D); Rijkswaterstaat's parser averages raw `NVT` readings even though a live 24 h mean parameter (`GEM24H`) exists; SYKE's public day values are morning nearshore reference readings (08:00, 20 cm below the surface) rather than 24 h means; and ARSO's daily value has no published mean-versus-hour statement. We therefore treat "daily mean" as a contract that holds for part of the panel and report the consequence: restricted to the five providers whose daily value is a documented provider daily mean, the development contrast is -5.7 points ($[-6.5, -4.9]$, 84 networks) and the reserved contrast -6.6 ($[-7.9, -5.3]$, 29 networks), against -6.4 and -7.1 for the full panel (`amendments/provider_contract_sensitivity/`). The sealed confirmation is USGS-only and so is entirely inside the documented stratum. The three SYKE stations remain flagged in `results/wrr_submission_ready/measurement_provenance_stations.cs` and are read as a sensitivity stratum, not as proven daily means. A *network* is one provider's group of stations on a common river system. It is the unit at which donor stations are available to one another, and it is the clustering unit for every interval reported here, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network and non-river objects — seas, lakes, canals, estuaries, deltas — are excluded rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day supplies no truth. Stations with fewer than four fully observed calendar years are excluded because the estimator protocol requires them. This leaves **146 networks, 457 stations and 5114 fully observed station-years**.

The same 457 stations contribute **2422 further station-years that are genuinely incomplete**. These carry no truth and are therefore not part of the scored population; they are the population a screening rule exists for. Section 4.12 reports overlap-rule decisions on that panel; the frozen certificate can also be applied through a truth-free `predict` interface (repository amendment `results/wrr_submission_ready/e2_certificate_predict/`).

Station coordinates were assembled for all 457 stations, from the providers' own catalogues, from the USGS NWIS site service for 195 United States gauges, and from

ARSO's public station feed, and daily 2 m air temperature was joined from NASA POWER for the whole panel, 1981 to 2026 (NASA Langley Research Center, 2026). For the United States subset, basin characteristics were joined from GAGES-II (Falcone, 2011) on the eight-digit gauge identifier, matching **140 of 204** eligible stations across 57 networks. Basin attributes are used to ask which rivers are hard (Section 4.7) and which rivers a screen keeps (Section 4.11); they are deliberately not used inside the screening rule, because they are unavailable for the other eight agencies.

Table 2 lists every quantity the analysis uses, what it means physically, where its value comes from, and whether it was calibrated. Networks — never station-years — were split into development and reserved, stratified on provider and on donor regime, with the seed recorded before any scoring. This gives **110 development and 36 reserved networks**. That reserved panel is a retrospective network holdout for the frozen v17 screen: 28 of 36 networks have prior development, descriptive or scoring use in earlier programme stages (`results/wrr_submission_ready/prior_access_ledger_reserved_networks.csv`). It is not an unused confirmation set for analyses developed after opening. A sealed confirmation was therefore recruited externally: 39 USGS HUC4 networks (91 stations; 18 HUC2 contexts; 8973 designed outage cases) that lie outside the v17/v18 development panel were ingested, hashed, and scored once with the frozen certificate (`results/wrr_submission_ready/p0_07_seal`). At 70% retention and 0.5 °C the pooled contrast is about −7.9 percentage points (clustered interval about $[-10.2, -5.8]$); the empirical arm is smaller but excludes zero (about −1.7 points, $[-3.0, -0.7]$). The panel is outside the development programme but it is not untouched by the wider programme, and we report that rather than assert otherwise: a repo-wide, provider-aware audit of every artifact carrying a station identifier finds 24 of the 39 networks appearing in earlier iterations, 15 networks with no outcome-bearing prior use and 4 with no prior use of any kind (`prior_access_audit_sealed_usgs.csv`; `AMENDMENT_prior_access_sealed_panel.md`). The result does not depend on the touched networks: restricted to the 15 outcome-clean networks the pooled contrast is −9.4 points ($[-12.4, -6.0]$) with an empirical-arm gain of −3.7 ($[-5.8, -1.6]$), and restricted to the 45 outcome-clean stations it is −8.3 ($[-10.4, -6.4]$). Those subsets are read from an already-sealed score and are a transparency diagnostic, not new confirmation. The panel remains USGS-only and uses provisional HUC4 groupings rather than agency river systems, which the multi-agency transport test below addresses. That test was pre-registered and run. Panels from Sweden (SMHI), England (UK Environment Agency) and France (Hub'Eau)

— 32 networks, 245 stations, 17 826 designed outage cases across boreal, temperate-oceanic and temperate-continental rivers, none of them used anywhere in the development programme — were sealed with input hashes and scored once with the same frozen configuration. The pooled contrast is -4.6 points ($[-6.5, -3.3]$), inside the pre-declared rule ($\Delta \leq -1$ point with a clustered interval excluding zero and at least two providers in the same direction), with each of the three new agencies negative and the quality-checked stratum agreeing (-6.2, $[-8.4, -4.5]$); the empirical arm is -1.9 ($[-3.3, -1.0]$). The panel carrying the pre-declared rule meets the 30-network, three-context design floor. Twenty-one additional Norwegian basins (NVE; individually never-used stations from a provider that took part in development) form a separate secondary stratum at -6.1 points ($[-8.7, -4.6]$), and extending the pooled estimate to them gives -5.2 ($[-6.7, -4.1]$) (`p0_07b_multi_agency_transport/verdict.jso`). The protocol records that the test was designed after the USGS score was known, so it is a transport test and not a pristine first confirmation; an earlier two-basin NVE probe is superseded by the secondary stratum (`p0_07_nve_probe/verdict.json`).

2.3 Gap design: two arms, and why both are needed

A case is one station-year under one gap realization. Two arms are used and reported separately, because they answer different questions and are not one population. The *empirical* arm samples real gap runs — length and start month — from the same network's own records and transplants them onto fully observed years, so that operational numbers reflect the geometry agencies experience. The *stratified* arm crosses gap length $\{14, 30, 60, 90, 120, 180, 270\}$ days with the overlap between the gap and the station's climatological peak window, so that the response surface is estimable rather than accidental. The design yields **51524 cases**, 40704 development and 10820 reserved.

Both arms transplant a mask onto a year whose values were observed, which assumes the mask is uninformative about the values it hides. Section 4.13 tests that assumption against the archive rather than asserting it. Gaps are placed inside one calendar year; failures that straddle New Year damage two annual statistics while hiding neither warm season, and the design cannot represent that case.

Each endpoint is scored against a tolerance equal to the same fraction, 0.47, of *its own* interannual standard deviation at the same stations. For the annual peak that scale is 1.06 °C, giving a primary tolerance of 0.50 °C; the scale falls monotonically with sup-

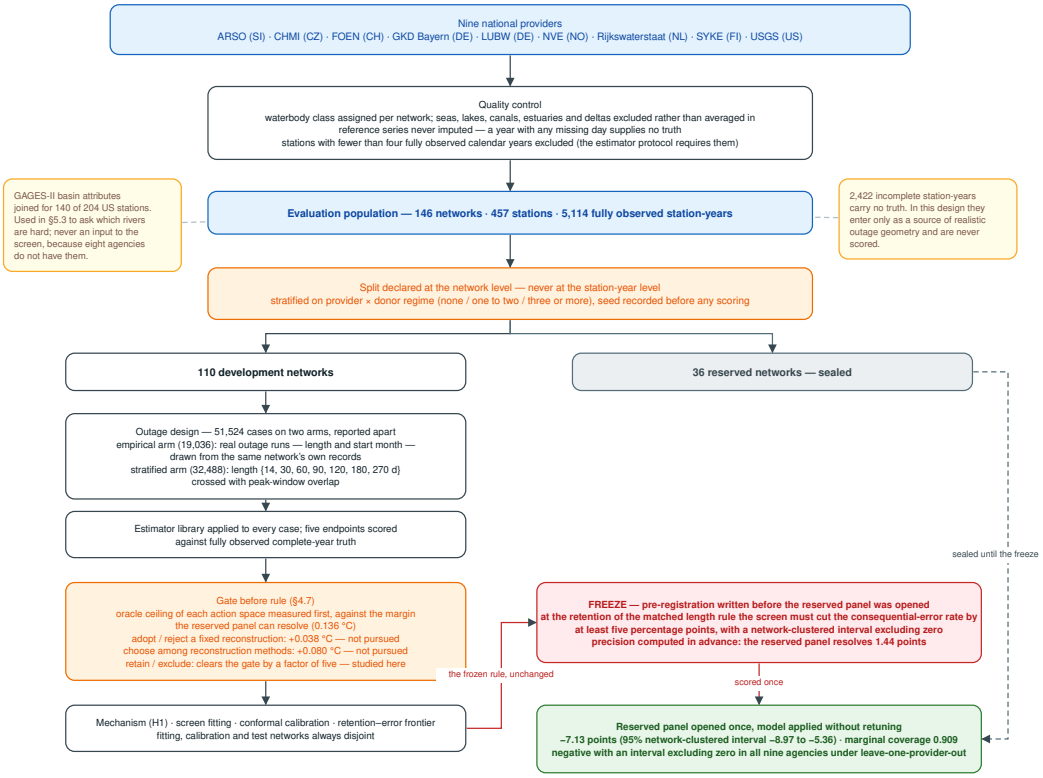


Figure 2. Study design and data flow from nine providers through quality control, the network stratified split, and the freeze. The v17 reserved panel is a retrospective holdout with mixed prior access; the sealed confirmation uses a separate USGS ingest outside the development panel, with a documented partial prior-use audit and a clean-subset diagnostic, plus a multi-agency transport panel with a labelled secondary NVE stratum.

Table 2. Quantities used in the analysis.

quantity	physical meaning	source	calibrated
h_q, hidden mass	share of the statistic's influence removed by the gap	mask and the station's own complete years	no
peak-window overlap	h_q for a 30-day extreme, from the climatological window	mask and training years	no
warm-season coverage	share of May–September days observed	mask	no
longest run, missing fraction	size of the gap	mask	no
observed-peak margin	how far the observed days already constrain the annual maximum	observed days and training climatology	no
donor count, coverage, correlation	how much synchronous anomaly information neighbours carry	training-period deseasonalised series	no
K, ridge penalty of the donor and air estimators	strength of shrinkage toward the seasonal mean	leave-one-year-out within station	yes, within station
conformal scale	multiplicative factor turning a predicted bound into a valid one	calibration networks, disjoint from fitting and test	yes, on networks
τ, tolerance	consequential error in the units of the statistic	0.47 of the endpoint's own interannual SD; swept 0.25–1.0	no, declared
basin attributes	elevation, slope, baseflow index, drainage area	GAGES-II, United States subset	no, not an input

aNo quantity in the first six rows reads a day the gap hides; that restriction is enforced in the feature constructor rather than by convention.

port width, from 1.18 °C at $k = 1$ to 0.55 °C at $k = 365$. The fraction is swept at 0.25, 0.47 and 1.0, and no conclusion about which axis wins is claimed unless it is invariant across the sweep. The reference series carries instrument error of roughly 0.1 to 0.2 °C, so the primary tolerance is about three times sensor accuracy; Section 4.13 perturbs the reference by 0.15 °C and repeats the analysis.

3 Methods

3.1 Reconstruction, and what each estimator is allowed to see

Every case is completed by each supported estimator: the statistic computed from recorded days alone (current practice), a day-of-year median climatology, a smoothed climatology, linear interpolation where both flanks exist, ridge regression on the visible subset of synchronous donors, and an air-driven ridge with lagged air temperature and harmonics. Estimators are fit leave-one-year-out within the station, which is what offline archive repair permits; Section 4.13 repeats the study under the stricter protocol in which only prior years are available. Support is declared per method: donor regression does not read a flanking observation, so requiring one would discard usable cases, and absent donor columns are never imputed but the model is refit on the visible subset.

One weakness in that library was worth removing before the comparison was run. Air temperature was available for 2.3% of development cases in an earlier version of this analysis, so the reconstruction this field actually uses was effectively absent, and 36 of the 110 development networks have no synchronous donor at all. With coordinates joined for all 457 stations and NASA POWER pulled for the whole panel, air forcing now covers 98.6% of development cases and 100% of reserved cases. The retrospective reserved holdout is not re-scored against the new forcing; Section 4.9 reports it as a development-panel sensitivity analysis.

The reconstruction layer is not the contribution of this paper and is kept deliberately simple. Its role is to make the comparison between conditioning variables fair, which requires only that the estimator paired with the incumbent rule is the strongest deployable one; Section 4.9 tests that requirement directly rather than assuming it.

3.2 The screening rule, and what it may not read

For each case the screen predicts an upper bound on the absolute error each supported estimator would produce, selects the estimator with the smallest bound, and emits a conformal interval on the statistic together with an admission decision at an analyst-chosen tolerance. Its inputs are the first six rows of Table 2. No input may read a day the gap hides, and that restriction is enforced in code rather than by convention.

Calibration is split conformal (Lei et al., 2018), and the risk-control framing follows the distribution-free line of Angelopoulos et al. (2022) with the caveat that selective observation is itself an existing problem statement (Bai et al., 2026); what is specific here is that selection happens among reconstructions chosen per station-year. Two properties matter. It is *selection-conditional*, because taking the smallest of several bounds is a winner's curse that per-estimator calibration cannot see, and *multiplicative*, because the error scale spans an order of magnitude between a gap that misses the peak window and one that covers it. Fitting, calibration and test networks are always disjoint.

3.3 Test protocol

Before any rule was fit we measured, on development cases, how much a rule with perfect foresight could win inside each candidate action space, and compared it with the margin the reserved panel can resolve. Two natural framings — adopt or reject a fixed reconstruction, and choose among reconstruction methods — have ceilings of $+0.038$ and $+0.080\,^{\circ}\mathrm{C}$ against a threshold of $0.136\,^{\circ}\mathrm{C}$, and were not pursued. Retain-or-exclude clears its gate by a factor of five and is the action space studied here. A decision experiment whose ceiling is below its own resolvable margin cannot produce evidence in either direction, and running one consumes a reserved panel for nothing.

Every interval resamples whole networks, because station-years within a network are not independent and reusing a station-year under several gap realizations does not create independent observations. The pre-registered primary criterion, frozen before the reserved panel was opened, is that at the retention of the matched length rule the screen must reduce the consequential-error rate by at least five percentage points with a network-clustered interval excluding zero. Precision was computed in advance: the reserved panel resolves 1.44 points, so the declared effect is 3.5 times the resolution.

Where several endpoints are compared, the direction of each contrast was predicted in advance from the form of the statistic, which is the defence against multiplicity. Three of those predictions were not met, and Sections 4.5 and 4.6 report them as failures rather than reshaping them into successes.

4 Results

4.1 Computing the peak on recorded days alone costs more than a year-to-year standard deviation

Current practice is expensive before any rule is proposed. On development cases, computing the annual peak from recorded days alone — the default once a station-year passes a completeness screen — gives a mean absolute error of **1.305 °C, or 1.23 interannual standard deviations** of the quantity being measured, with 24.9% of station-years wrong by more than 1.0 °C. Substituting a day-of-year climatology over the gap gives 0.402 °C, the best supported reconstruction 0.225 °C with air forcing available for the whole panel (0.242 °C without it), and an oracle over the estimator set 0.130 °C. The error is therefore dominated not by which station-years are kept but by whether the record is repaired at all, which fixes the size of the question the rest of the paper addresses: given that a year has been repaired, which repaired years can be trusted.

4.2 The calibrated bound is honest enough to interpret

Everything that follows uses a predicted error bound to rank station-years, so the bound is checked before it is used. On the reserved networks the nominal 90% interval covers 90.9% of cases marginally, inside the pre-declared band of 0.87 to 0.93, and 86.3% to 94.9% within every pre-declared stratum (Figure 3a). Median widths adapt with the quantity they are meant to track, from 0.06 °C when the peak window is untouched to 1.87 °C when most of it is hidden.

Those strata pool across networks, and conformal validity assumes an exchangeability that fails between river networks. Coverage inside individual reserved networks has mean 0.904 and median 0.913 but a minimum of 0.778, with five of 36 networks below 0.85. The pooled interval is an honest statement about a population of networks and is *not* a guarantee at a single new site; calibrating on the per-network conformal quantiles at the 0.9 level puts no network below 0.85 for a 1.5-fold widening. The screening

results below are identical under every calibration to four decimal places, because the rank order the screen uses is a monotone transform of the bound, so a reader who distrusts the intervals need not distrust the frontier.

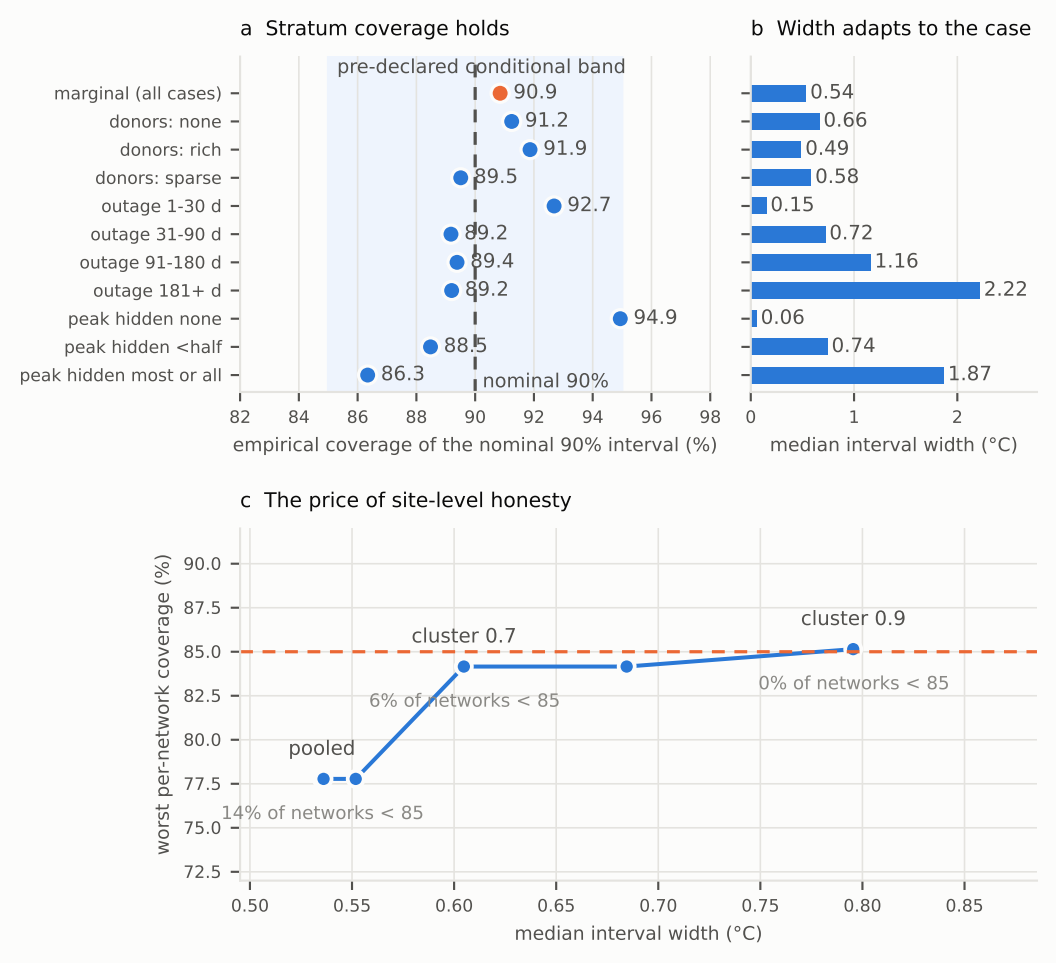


Figure 3. Calibration diagnostics on held-out cases: coverage by donor-regime stratum at the nominal 0.90 level, with the 0.90 line (a); interval width (°C) against the share of the peak window the gap hides, with network-clustered bands (b); and the width price of a site-level rather than pooled guarantee (c). Development cases unless a panel is labelled otherwise; the site-level comparison is exploratory because its level was chosen on the opened reserved panel.

4.3 The share of the support a gap hides orders every endpoint, and the ordering reverses

Table 3 reports, for five endpoints, how well each candidate property of the gap discriminates a consequential error. The ordering reverses between extremes and averages,

Table 3. Discrimination of a consequential error by axis (AUC), development cases, each endpoint at 0.47 of its own interannual standard deviation.

endpoint	peak-window overlap	warm-season coverage	outage length	AUC difference (95% CI)
hottest 7-day mean	0.823	0.824	0.752	+0.072 (0.063, 0.096)
hottest 30-day mean	0.819	0.820	0.750	+0.070 (0.061, 0.098)
amplitude	0.758	0.792	0.768	+0.024 (0.015, 0.037)
annual mean	0.672	0.818	0.845	−0.027 (−0.036, −0.018)
coldest 30-day mean	0.447	0.619	0.803	−0.184 (−0.228, −0.156)

aThe difference is AUC(best timing axis) minus AUC(outage length). All five intervals exclude zero and the ordering reverses between the two regimes.

every clustered interval excludes zero, and the winning axis is unchanged at tolerances of 0.25, 0.47 and 1.0 interannual standard deviations. The seven-day peak has the largest timing advantage of any endpoint, which the support account predicted before it was measured because a shorter extreme is more localised in time; peak-window overlap has an AUC of 0.447 for the cold extreme, because hiding the summer peak carries no information about the winter minimum. The last row is not evidence that cold extremes are a length-governed exception, and reading it that way would be a mistake: the two support axes in the table are defined on the warm season. Built on the endpoint's own season — a climatological coldest-30-day window and the observed share of the cold months — the support axis beats the length rule for the coldest 30-day mean on the same development cases (AUC 0.876 and 0.897 against 0.819; clustered error-rate gaps −0.0050 and −0.0059, intervals excluding zero), and adds to length in a grouped out-of-fold model (AUC 0.905 against 0.813 for length alone). What the table shows is that the support must be defined on the season the statistic lives in; it does not license a warm-season axis for a cold endpoint. That cold-endpoint analysis is exploratory: it uses the development panel and does not touch the sealed confirmation.

4.4 The transition is continuous, and a fixed narrow window turns over near a three-month support

Table 3 evaluates the question at two points and reads the gap between them as a dichotomy. Sweeping the hottest-k-day mean of Equation (1) from $k = 1$ to $k = 365$ evaluates it everywhere (Table 4, Figure 4a). The advantage of a fixed 30-day peak window — the axis the screen actually uses — is flat at about +0.075 AUC while the estimand's support is a month or less, falls to +0.050 at two months and to +0.002 at three, and is negative beyond that. The 90-day grid point is statistically indistinguishable from zero (95% interval −0.018 to +0.026, positive in 55% of network-clustered replicates), and the first clearly negative grid point is 182 days, so the crossing is only located between 60 and 182 days: a bootstrap of the whole curve puts its 95% interval at $[77, 107]$ days and it moves between 85 and 102 days across the tolerance grid, while the opened reserved panel's interval ($[104, 136]$) overlaps it rather than displacing it. We therefore report a range and not a day count. The same sweep supports a second, different statement: a conditioning window dilated around the estimand's own support keeps its advantage over whole-year completeness out to about 359 days (95% interval 351 to 364, and it fails to cross at all in 7% of replicates), ending only at the trivial $k = 365$ coincidence, and the advantage is a hump that peaks near 60 days rather than decaying monotonically. The mid-range crossing is thus a property of holding the conditioning window narrow and fixed, not of the support idea itself. Nor is one window enough across the continuum: a single 60-to-90-day window is within 0.0004 AUC of each endpoint's own optimum up to a 90-day support, but costs 0.083 AUC at the annual-mean end. These continuum numbers are exploratory analyses developed after the reserved panel was opened; they motivate the screening axis but are not the sealed USGS confirmation reported with the data description. The two regimes of Table 3 are useful ends of that exploratory scale, not a proven universal dichotomy.

The mechanism behind that scale is measurable rather than only plausible. Binning development cases on both properties simultaneously and measuring the growth of error along each with the other averaged out, the hottest 30-day mean grows 3.5× along the missing-days axis and 9.6× along the peak-overlap axis, while the annual mean grows 10.2× and 1.5× respectively: the two ratios swap by roughly an order of magnitude in each direction. For the peak, once the support is in the model, gap length adds nothing measurable — its permutation importance is −0.004 against 0.194 for peak overlap

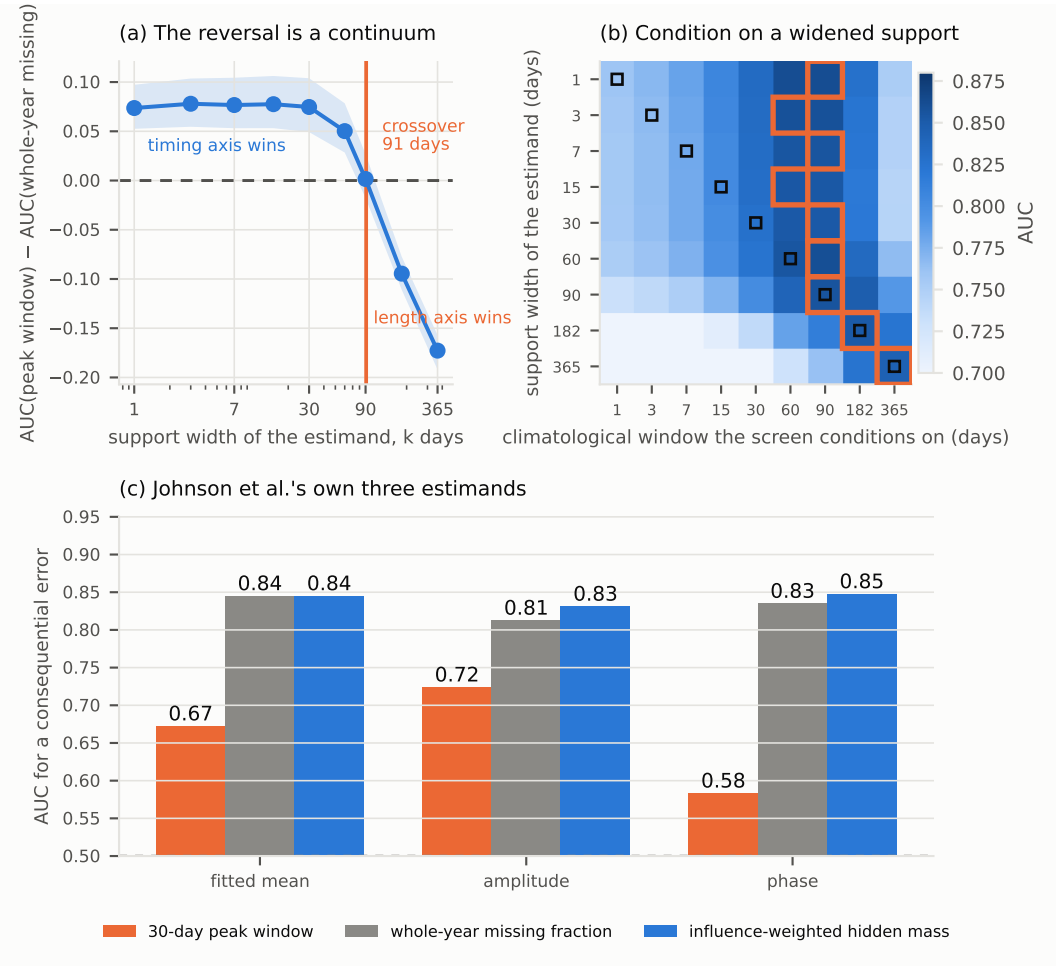


Figure 4. Exploratory continuum on the opened panel (not the sealed USGS confirmation). (a) Advantage of a fixed 30-day peak window over whole-year completeness versus support width, with network-clustered intervals; the crossing is located between 60 and 182 days (bootstrap 95% interval $[77, 107]$ days) and the 90-day grid point is indistinguishable from zero. (b) AUC for every estimand against every conditioning window. (c) Sine-fit parameters of Johnson et al. (2021), by axis.

Table 4. The governing axis as a function of the support of the estimand, development cases.

support width k (days)	interannual SD (°C)	AUC, peak window	AUC, missing fraction	difference (95% CI)
1	1.18	0.825	0.751	+0.074 (0.053, 0.097)
3	1.17	0.824	0.746	+0.078 (0.055, 0.104)
7	1.16	0.823	0.746	+0.077 (0.053, 0.104)
15	1.12	0.821	0.743	+0.078 (0.053, 0.106)
30	1.08	0.819	0.744	+0.075 (0.049, 0.104)
60	1.02	0.815	0.765	+0.050 (0.028, 0.078)
90	0.94	0.792	0.790	+0.002 (−0.018, 0.026)
182	0.74	0.728	0.823	−0.095 (−0.111, −0.077)
365	0.55	0.672	0.845	−0.173 (−0.192, −0.157)

aThe difference is AUC(30-day peak window) minus AUC(whole-year missing fraction), with a network-clustered 95% interval. Every interval excludes zero except the one at $k = 90$, which is where the crossing is.

(Figure 5c) — and nested feature sets grouped by network rise from AUC 0.742 with length alone to 0.845 adding the support, 0.858 adding the observed-peak margin and 0.884 adding donor quality (Figure 5b).

Two qualifications belong with that. Length is not irrelevant: holding peak overlap at zero, the consequential-error rate still rises from 0.6% at 1 to 30 days to 8.7% at 121 to 200 days. And in a linear score that already contains warm-season coverage the length term takes a small negative weight, which is collinearity — the two correlate at −0.86 — and not evidence that longer gaps are safer. That correlation is also why a timing-versus-length labelling is wrong: warm-season coverage is a count of missing days restricted to a window, which is a length measure with a domain restriction, and it is the domain restriction and not the timing that does the work.

Donor information enters the residual through quality rather than quantity. Halving the donor set by training-period correlation costs +0.084 °C (95% interval 0.064 to 0.110) when the comparison is re-run with the donor count matched exactly between the two arms, against +0.082 (0.061 to 0.112) under the original construction in which the

paired cases can differ by up to seven donors (34% of cases). Replacing donor anomalies with phase-shuffled surrogates while preserving each donor's seasonal cycle costs +0.234 °C (0.149 to 0.299), so donors supply synchronous anomaly information and not merely a seasonal shape.

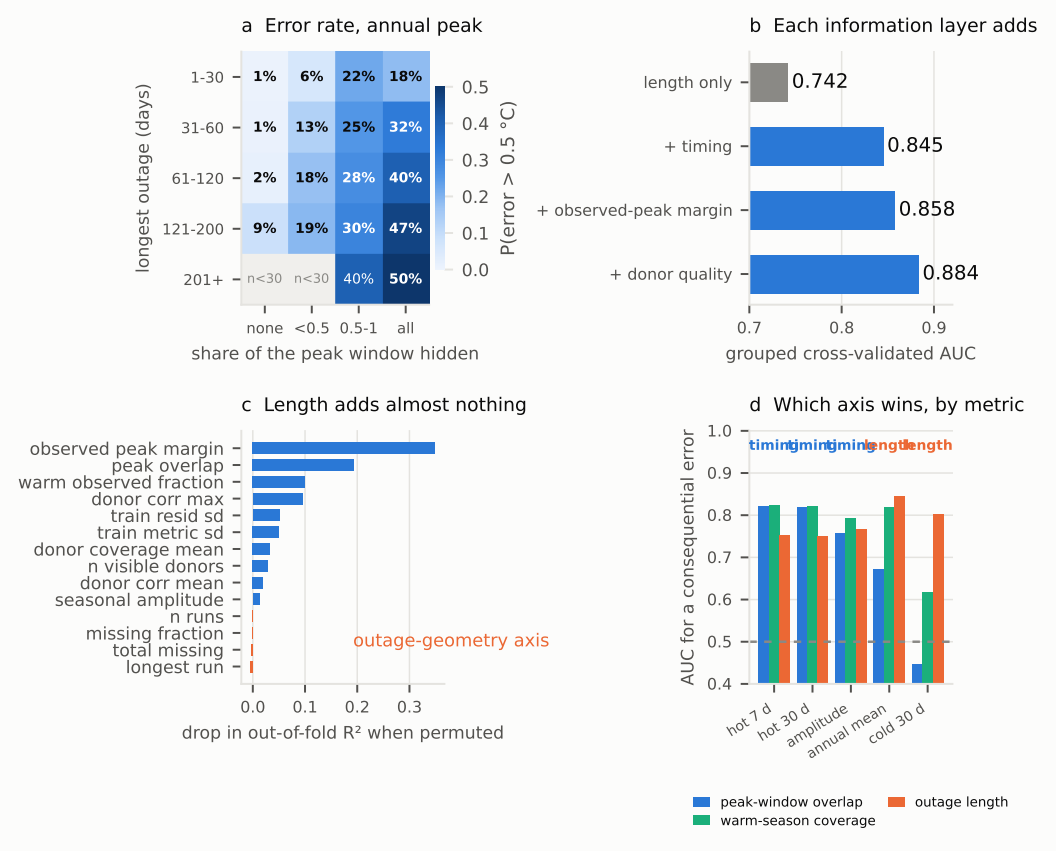


Figure 5. Mechanism on development cases (n = 40 704 cases, 110 networks; error bars are network-clustered 95% intervals unless a panel says otherwise): the length-by-overlap reversal (a), nested feature sets with out-of-fold AUC (b), permutation importance of the decision-time features (c), and the endpoint dependence of the winning axis (d).

The sine-fit parameters also connect to a physical reading that is worth keeping separate from the statistical one: air–water phase and amplitude carry information about groundwater and riparian influence (Hare et al., 2023), and a support-conditioned screen improves the error of those parameters without claiming to identify the process behind them.

Table 5. Which climatological window a screen should condition on, development cases.

estimand k (days)	best window (days)	AUC there	AUC on own support	gain	support wander (days)
1	90	0.862	0.760	+0.102	16.7
3	60	0.858	0.766	+0.092	16.3
7	90	0.855	0.778	+0.077	16.5
15	60	0.853	0.802	+0.051	15.3
30	90	0.852	0.827	+0.025	13.7
60	90	0.860	0.855	+0.005	9.4
90	90	0.856	0.856	0	7.4
182	182	0.851	0.851	0	6.9
365	365	0.844	0.844	0	—

aThe diagonal is the estimand's own support; the last column is the median year-to-year circular standard deviation of the realised window's centre.

4.5 A support must be located before it can be conditioned on, so the useful window is wider than the statistic

Two predictions declared for this round were not met, and both failures have one cause. We predicted that the axis restricted to the estimand's own support would be at least as good as any hand-chosen axis at every k, and that its advantage would decay monotonically. Neither holds: the advantage is a hump peaking at a 60-day support, and for supports of 15 days or less the fixed 30-day peak window beats the estimand's own support, by 0.064 AUC at $k = 1$.

Table 5 gives the reason. The gain from conditioning on a window wider than the statistic tracks the year-to-year wander of the realised window's location almost exactly: it is +0.102 AUC at $k = 1$, where the climatological window is the worst estimate of where this year's window will fall and the wander is 16.7 days, and vanishes at $k \geq 90$, where the wander has fallen to 7.4 days and the support is wide enough to contain its own location error. A screening rule therefore does not condition on the support; it conditions on the support *dilated by the uncertainty in the support's location*, and a dilation of roughly two location standard deviations on each side is what the data choose.

One consequence is practical. AUC at the best conditioning window is between 0.85 and 0.86 for every estimand from the annual maximum to the half-year mean, so an agency needs one climatological window of 60 to 90 days rather than one per metric. A second is uncomfortable: the screen deployed below conditions on a 30-day window and is therefore leaving 0.025 AUC on the table for its own endpoint.

4.6 On the estimands the standing guidance was derived for, the guidance is right

Johnson et al. (2021) fit a sine wave by linear regression and take a mean, a phase and an amplitude; they do not compute a hottest 30-day mean. Reproducing that estimator on our panel puts the two accounts on the same footing for the first time (Table 6, Figure 4c). A completeness threshold is the correct axis for the fitted mean, at 0.845 against 0.672 for peak-window overlap, and is within 0.02 AUC of the best axis for the phase. The ordering is the same under the authors' own protocol, in which the sine is fitted to whatever days are present with no repair at all: peak overlap 0.667, 0.743 and 0.590 against whole-year missingness 0.902, 0.842 and 0.898 for the mean, amplitude and phase, at base rates of 32% to 34% consequential against 10% to 12% after repair.

Two reporting rules come out of this endpoint. Phase is only identifiable when the fitted amplitude stands above the noise: the standard error of a three-parameter sine fit is $\sigma\sqrt{2/n}/A$ radians, so requiring a 95% interval no wider than the phase tolerance used here gives $A/\sigma \geq 4.5$, which retains 70.7% of cases and leaves the comparison above unchanged (+0.0126 against +0.0127 on all cases); under the authors' own unrepaired protocol the defect is active rather than latent, with 0.10% of cases below $A/\sigma = 1$ carrying a median phase error of 60 days against 91 days for a random phase. The rule is an identifiability floor, not a quality guarantee: the error rises again above $A/\sigma \approx 8$. Second, the fitted mean and phase are the regime the standing guidance was derived for, and on them it is right.

The third declared prediction failed here. We expected that for the phase, whose influence is largest a quarter-year from the peak, two 61-day windows around the spring and autumn limbs would beat both peak overlap and gap length. They beat peak overlap decisively (+0.249 AUC, 95% CI 0.231 to 0.270) and they do not beat gap length (0.831 against 0.836). Weighting every day by the influence function of Equation (3) in-

Table 6. Axis comparison on the sine-fit parameters (AUC), development cases, each at 0.47 of its own interannual standard deviation.

estimand	peak window	missing fraction	outage length	influence-weighted	best axis
fitted mean	0.672	0.845	0.845	0.845	outage length
amplitude	0.724	0.812	0.815	0.831	influence-weighted
phase	0.583	0.835	0.836	0.847	influence-weighted

stead recovers the advantage: 0.847, which beats the whole-year missing fraction by +0.013 (0.009, 0.017) and peak overlap by +0.265 (0.249, 0.285).

4.7 Topography, not geology, makes a river hard

The account so far is a property of estimators and estimands and would hold for any time series. For the 140 United States stations matched to GAGES-II across 57 networks, we regressed donor synchrony and completed-year error on basin attributes with the expected direction of each effect declared before the regressions were run (Figure 6). Site elevation and basin slope both reduce donor synchrony (Spearman $\rho = -0.29$, 95% CI -0.46 to -0.08, and $\rho = -0.31$, -0.51 to -0.07), and site elevation raises both completed-year error ($\rho = +0.24$, 0.06 to 0.43) and the consequential-error rate ($\rho = +0.25$, 0.08 to 0.43).

The groundwater prediction taken from the literature is not reproduced, and the reason is instructive. Johnson et al. (2021) report that filling can harm groundwater-influenced sites, which predicts larger completed-year error at a high baseflow index. Baseflow index does reduce donor synchrony in the predicted direction, from 0.70 in the lowest tertile to 0.59 in the highest, although the rank correlation interval includes zero ($\rho = -0.19$, -0.42 to 0.10). But the error moves the other way: splitting at the median baseflow index, high-baseflow stations are $-0.043\,^{\circ}$C easier than low-baseflow stations (95% CI -0.081 to -0.004). Across baseflow tertiles the mean annual peak falls from 22.7 °C to 18.1 °C ($\rho = -0.47$, -0.60 to -0.26) and its interannual standard deviation from 0.98 to 0.76 °C (Figure 6c), so two effects act in opposite directions and the second is larger.

Regulation does not make station-years harder for this endpoint: reference and non-reference basins differ by $+0.004\,^{\circ}$C (95% CI -0.051 to $+0.093$). Catchment attributes

explain a modest share of the variation, a standardised linear model reaching $R^2 = 0.22$ for donor synchrony and 0.13 for completed-year error, and no attribute predicts how much a screen gains: every correlation between a basin attribute and the per-station screening benefit has an interval containing zero. Whether an attribute predicts what a screen *keeps* is a different question, and Section 4.11 answers it differently.

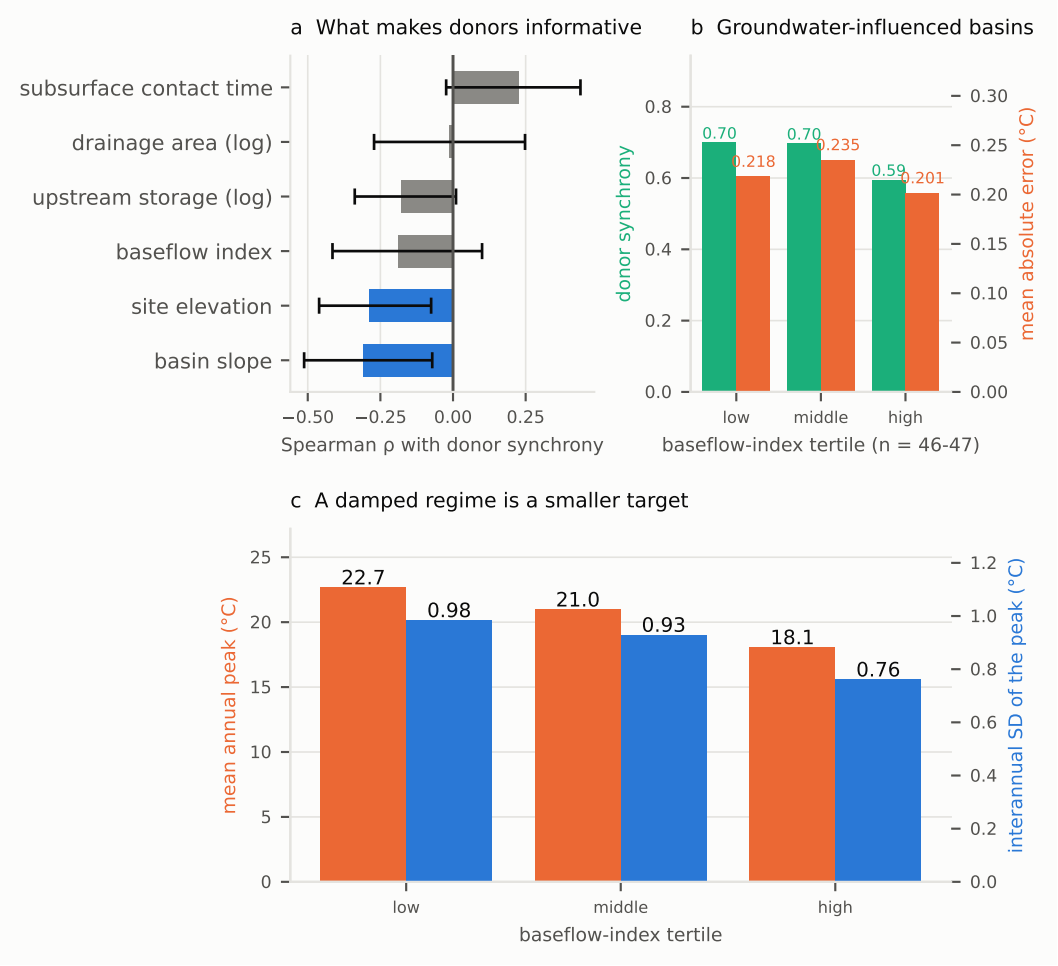


Figure 6. Catchment attribution on the United States subset (140 of 204 stations, 57 networks, GAGES-II attributes; points are stations, bands are network-clustered 95% intervals): donor synchrony against site elevation (a), completed-year absolute error (°C) against elevation (b), and the decomposition of the baseflow-index effect (c). These are attribute associations on one country, not a geology law.

Table 7. Reserved panel at 70% retention, across tolerances.

screen	$> 0.25\,^{\circ}\mathrm{C}$	$> 0.5\,^{\circ}\mathrm{C}$	$> 1.0\,^{\circ}\mathrm{C}$	under by $> 1\,^{\circ}\mathrm{C}$
outage-length rule	17.2%	9.7%	3.4%	2.0%
peak-overlap rule	9.2%	4.2%	1.2%	0.7%
calibrated screen	7.1%	2.5%	0.4%	0.3%
random	23.9%	14.4%	5.6%	3.7%

aThe last column is the share of retained station-years whose annual peak is underestimated by more than $1\,^{\circ}\mathrm{C}$.

4.8 Conditioning on the support cuts the error rate at equal retention, and three variables carry most of it

At 70% retention on development cases, the share of retained station-years with a peak error above $0.5\,^{\circ}\mathrm{C}$ is 2.9% for the calibrated screen, 5.6% for a rule on peak overlap alone, 9.2% for the gap-length rule and 15.0% for random retention (Figure 7a). On the 36 reserved networks scored as a retrospective network holdout (mixed prior access; see the prior-access ledger), the screen gives 2.5% against 9.7% for the length rule (Figure 7b): a contrast of -7.13 points, 95% network-clustered interval -8.97 to -5.36, against the five-point effect declared for that freeze. Both declared conditions are met on that holdout, in every donor regime (-6.6, -6.7, -7.2, all excluding zero). An independent sealed USGS confirmation recovers a similar pooled contrast (Section 2.2). The holdout result is not carried by the largest agencies alone: case-equal -7.01, provider-equal -6.29, network-equal -5.82, station-equal -5.54 points, with a median across individual reserved networks of -5.0, 83% of networks favouring the screen, and a worst network of $+2.8$.

Placed at the operating points the published guidance implies rather than at an arbitrary retention: at the seven-week unfilled limit (49 days, 49.2% retention on the reserved panel) the length rule leaves 5.2% and the screen 0.3%; at the twenty-week imputed limit (140 days, 85.1% retention) the length rule leaves 11.3% and the screen 6.9%. The second of those is more permissive than an earlier version of this analysis assumed — it admits 85% of cases rather than 74% — and the screen's advantage there is 4.4 points rather than 6.2.

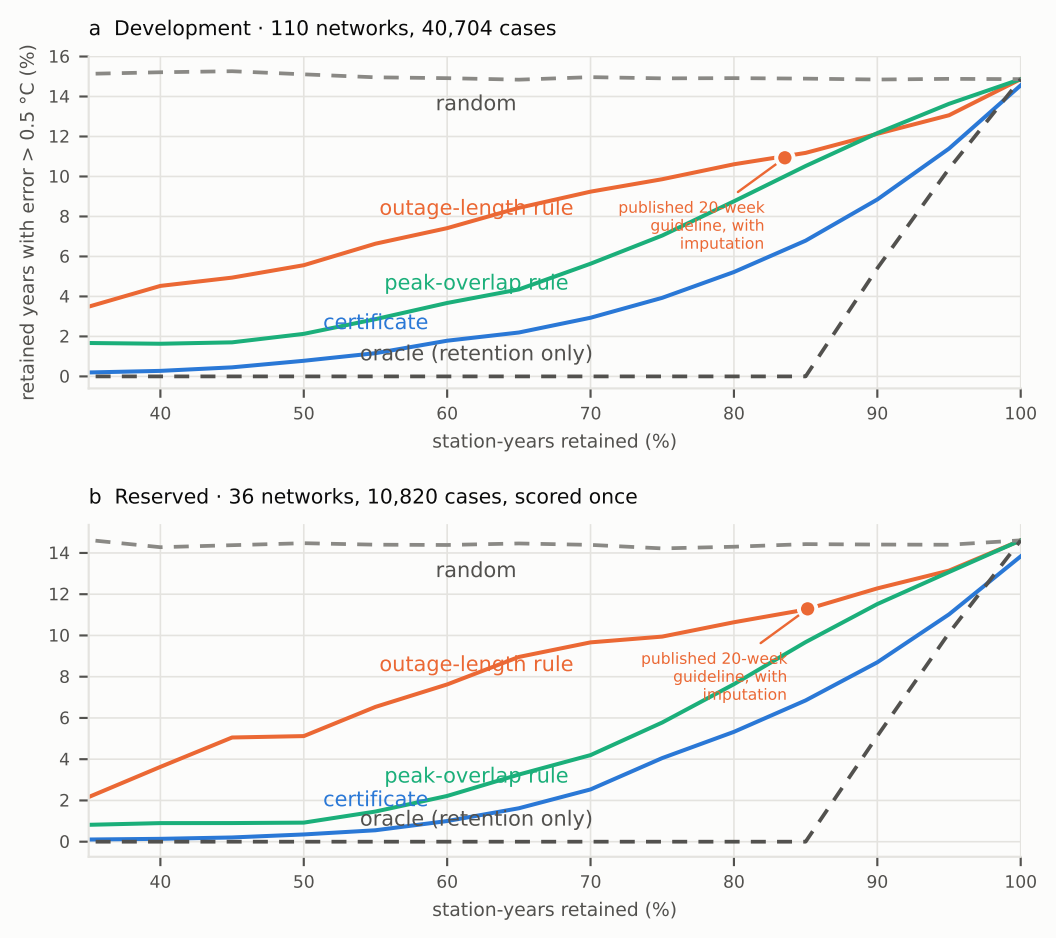


Figure 7. Retention-error frontier on development cases (a) and on the reserved panel (b).

Table 8. Component ablation, development cases at 70% retention, each variant re-run through the full out-of-fold pipeline.

screen	error rate	points vs the length rule
outage-length rule (incumbent)	9.24%	—
three-variable logistic regression	5.53%	−3.71
peak-overlap alone, no fitting at all	5.62%	−3.62
three-feature gradient boosting	4.89%	−4.35
geometry and timing only (8 features)	4.86%	−4.38
drop donor features	3.78%	−5.46
drop the observed-peak margin	3.13%	−6.11
full screen	2.93%	−6.31
full screen, screening only (no method choice)	3.11%	−6.13

Most of the benefit does not require the calibrated machinery (Table 8). Fifty-nine per cent is available from a logistic regression on three variables and 57% from peak-window overlap with no fitting at all; method selection contributes 0.2 points. Fitted on development networks and applied unchanged to the reserved panel, the deployable rule is

$$\text{risk} \;=\; \text{logistic}\big(-1.302 + 2.070\,x_1 - 2.412\,x_2 - 0.00187\,x_3\big), \tag{4}$$

where x_1 is the share of the station's climatological hottest-30 window the gap hides, x_2 the share of May-to-September days that remain observed, and x_3 the length in days of the longest consecutive gap. It gives 4.2% against the length rule's 9.6% at 70% retention, a contrast of −5.44 points (−6.85, −3.96), and cuts severe underestimation of the peak from 1.9% to 0.6%.

4.9 The incumbent is not handicapped, including by the absence of the field's own reconstruction

A screening comparison can be rigged by pairing the incumbent with a weak reconstruction, so the incumbent is paired here with every reconstruction in the study, including one nobody can deploy (Table 9).

Table 9. Development cases at 70% retention, with air forcing available throughout.

screen	reconstruction	error rate above 0.5 °C
length rule	information-conditioned policy (used throughout)	9.00%
length rule	always library air ridge, $\alpha = 1.0$, else climatology	15.16%
length rule	always faithful Beaufort et al. (2022) climate correction	12.76%
length rule	always donor ridge, else climatology	9.57%
length rule	always smooth climatology	19.79%
length rule	oracle over methods, not deployable	3.62%
calibrated screen	deployable estimators it selects itself	2.87%

The screen beats the incumbent armed with a reconstruction nobody can run, and the reconstruction paired with the incumbent throughout is the strongest of the deployable options rather than the weakest. Air-driven reconstruction is reported twice, because the earlier label was wrong: air-temperature regression remains a workhorse for data-sparse basins (Sanikhani et al., 2026; You et al., 2026), and the library's air ridge with a fixed penalty ($\alpha = 1.0$, seven lags, whole-year fit, penalised) was described as "in the manner of Beaufort et al. (2022)" but shares only its input with that paper, and is now labelled as what it is. Reimplementing the cited method faithfully — per-station OLS of summer water temperature on a right-aligned moving average of air temperature, 2–10 day windows chosen per station by R^2, fitted on July–August — gives 12.76% against the library ridge's 15.16% (paired difference −2.40 points, $[-3.20, -1.62]$, 1200 network-clustered draws), and both remain worse than per-case selection at 9.00% (`amendments/faithful_baseline/`). The axis conclusion does not depend on the estimator library: the timing-minus-length AUC contrast is +0.070 with the old forcing and +0.081 with the new. The screen uses the new information, selecting the air-driven estimator for 20.0% of cases against 0.07% before, and donor regression for 53.1%.

One result here was not expected: the *fixed* information-conditioned policy gets slightly worse at donor-less stations when air becomes available, from 0.340 to 0.364 °C.

Table 10. Decomposition by outage arm.

panel	arm	cases	base rate	length rule	screen	contrast (95% CI)
development	empirical	14995	7.4%	2.52%	0.38%	−2.13 (−3.45, −1.26)
development	stratified	25709	19.3%	14.58%	6.17%	−8.39 (−9.25, −7.47)
reserved	empirical	4041	4.2%	0.74%	0.11%	−0.71 (−2.36, +0.05)
reserved	stratified	6779	20.8%	16.59%	7.78%	−8.68 (−10.46, −6.51)

aBase rate is the consequential-error rate with no screening; the two rule columns are at 70% retention. The declared five-point effect is met on the pooled and stratified populations and not on the empirical arm.

4.10 The gain disappears under the gap calendar these networks keep, because long gaps begin in autumn

The frozen holdout test was run on the pooled case set, which mixes two arms whose base rates differ by an order of magnitude, and they must be reported apart (Table 10). The declared five-point effect is met on the pooled and stratified populations and *not* on the reserved empirical arm, where the interval includes zero. The holdout headline −7.13 points is therefore an effect on a mixture that includes a designed arm. The sealed USGS confirmation likewise finds a much smaller empirical-arm gain than the pooled contrast, so neither number should be quoted as next year's agency experience under natural outages alone.

Why the empirical arm is null is itself a result. Across the 6472 real interior gaps in this archive, gaps shorter than 30 days and gaps of 30 to 89 days are close to uniform in start month ($\chi^2 = 4.0$ on 11 df, $p = 0.97$ for the 30-to-89-day class). Gaps of 90 days or more are not: **72.3% of them begin between September and January and 17.9% between March and July** ($\chi^2 = 86.1$, $p = 9.5 \times 10^{-14}$; Figure 8a). The gaps long enough to destroy a summer peak are seasonal shutdowns, and they start after the warm season has been recorded, so little is at risk and little can be screened out: 4.2% of reserved empirical cases exceed the tolerance with no screening at all.

A counterfactual that holds gap lengths fixed and moves only start month is reported in Figure 8b as an exploratory amendment after correcting year-end truncation in the placement code (`results/wrr_submission_ready/e1_maintenance_recompute/`).

Those percentage-point swings are not sealed confirmation and are not used as agency recommendations.

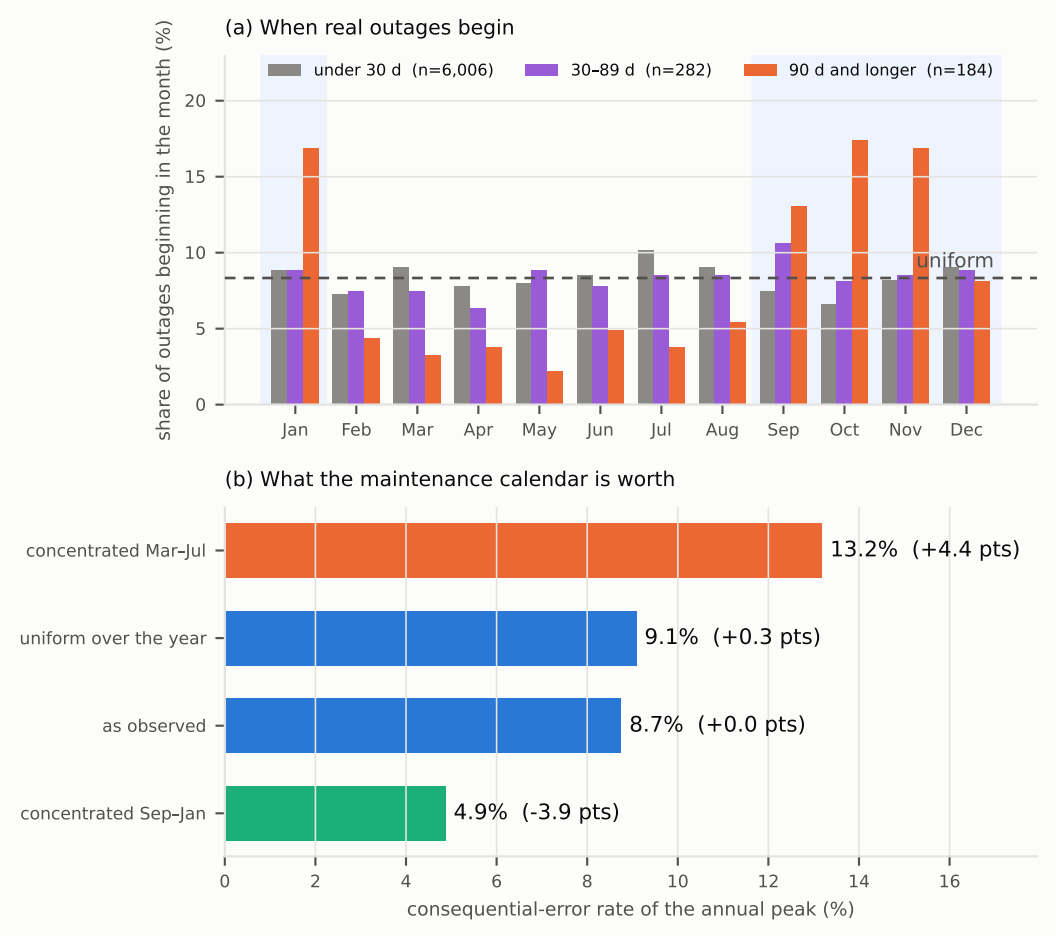


Figure 8. (a) Start month of 6472 real interior gaps by length class; long outages begin in autumn and winter (descriptive). (b) Exploratory maintenance-calendar counterfactual after multi-year placement correction; not a verified operational benefit.

4.11 Conditioning on difficulty removes mountain catchments unless the threshold is applied within strata

A rule that lowers the average error by discarding hard station-years has bought accuracy with representativeness, and in this application the hard sites are not a random subset: Section 4.7 showed they are high-elevation, steep, donor-poor catchments, which are the cold-water headwaters the summer endpoint exists to describe.

Table 11. Error against representativeness, reserved panel at 70% retention.

rule	error rate	ρ with elevation	donor spread	retention SD
outage-length rule	8.81%	−0.10	4.5 pts	0.075
peak-overlap rule	3.75%	+0.16	2.4 pts	0.104
calibrated screen, global threshold	2.53%	−0.27	4.3 pts	0.149
screen, balanced within donor regime	2.52%	−0.27	0.1 pts	0.148
screen, balanced within elevation tertile	2.85%	+0.08	4.6 pts	0.148
screen, balanced within both	2.84%	+0.07	0.1 pts	0.144

aRetention SD is the standard deviation of per-station retention across stations; a larger value means the rule treats stations more unevenly.

At 70% retention the calibrated screen's per-station retention correlates with site elevation at Spearman $\rho = -0.26$ ($p = 0.011$) on development and -0.27 ($p = 0.077$) on the reserved panel, while the incumbent length rule is elevation-neutral at $\rho = -0.05$ ($p = 0.60$) and -0.10 ($p = 0.52$). On the reserved panel the screen retains 77.7% of station-years in the lowest elevation tertile and 66.9% in the highest, a spread of 10.8 points against 1.7 for the length rule (Figure 9b). By donor regime the same pattern appears on development cases: 57.1% at donor-less stations against 72.5% at donor-rich ones, a 15.4-point spread, while the length rule tilts the other way.

The distortion is cheap to remove (Table 11). Applying the screen's threshold within stratum rather than globally costs 0.31 points of consequential-error rate on development cases and 0.30 on the reserved panel and flattens both distortions. Stratifying on donor regime alone does not fix the elevation tilt and stratifying on elevation alone does not fix the donor tilt, so an agency that cares about both must declare both.

4.12 On records that really have gaps, the two rules disagree on one year in six

Every number above is computed on a complete station-year with a synthetic hole punched in it, which is the only way to have a truth. The same archive contains 2422

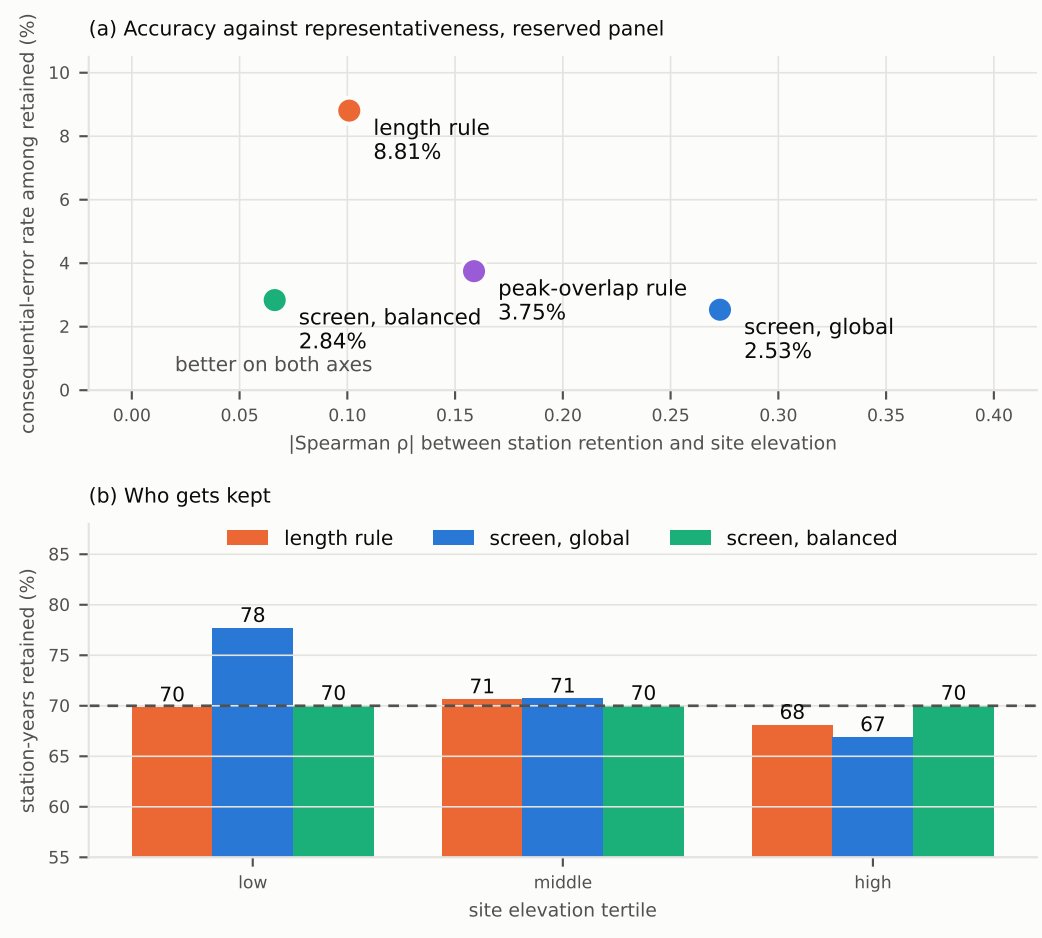


Figure 9. (a) Consequential-error rate against the correlation between station retention and site elevation, reserved panel. (b) Retention by elevation tertile under three rules.

Table 12. Archive composition under each admission rule, on the 2422 real incomplete station-years.

rule	admitted	any day hidden	half hidden	wholly hidden
repair everything, no screening	2422	696	261	186
length $\leq$ 140 d, with imputation	2221	507	82	17
length $\leq$ 49 d, unfilled	2016	437	45	2
peak window $\leq$ 10% hidden	1967	241	0	0

aBoth published limits are those of Johnson et al. (2021). The last three columns need no truth: they count admitted station-years in which any day, more than half, or all of the climatological peak window is unobserved. A wholly hidden climatological window is a proxy for a wholly unobserved realised peak, with a measured positive predictive value of 0.74 on sealed designed cases (Section 4.13).

genuinely incomplete station-years, a third of the total, which carry no truth and have never been decided. Three things can be asked of them without inventing one (Figure 10).

They are not a marginal population. The median incomplete year is missing 9 days and the 90th percentile 128; 28.7% of them have at least one day of the station's climatological peak window missing, and 261 have more than half of it missing.

The two rules disagree on a sixth of them. At the seven-week unfilled limit the length rule admits 83.2% and a peak-window rule at 10% hidden admits 81.2% — near-equal retention, which is the comparison this paper insists on — and they **disagree on 15.1%** of station-years: 207 are admitted by the incumbent and rejected on the support, and 158 are rejected by the incumbent and admitted on it. At the twenty-week imputed limit the incumbent admits 91.7% and the disagreement is 11.6%, of which 267 are in the dangerous direction and 13 in the wasteful one. The disagreement is present in every provider with more than 20 incomplete station-years, from 11.1% to 25.7%.

What the disagreement costs can be counted without a truth (Table 12). The published guideline agencies are most likely to apply, the twenty-week limit that exists because imputation is assumed, admits 17 station-years whose *climatological* peak window was never observed at all and 82 whose peak window is mostly gone. That count is a proxy and its error rate is measurable: on the sealed designed cases, where the realised peak is known, a wholly hidden climatological window carries a positive predictive value of 0.74 (network-clustered interval 0.71 to 0.76) for a wholly hidden *realised* peak, with sensitivity 0.87 and only 3.1% of proxy-clear cases hiding the realised peak. Transported

to the 17 flagged station-years, that implies about 12 to 13 whose realised annual peak was wholly unobserved and about 4 whose proxy fired while the realised peak was visible. A risk model fitted on development cases from decision-time features only, and checked for calibration on the reserved panel first (largest decile miscalibration 6.2 points, and under 1.6 points in nine of ten deciles), predicts a consequential-error rate of 0.98% among the years the unfilled limit admits, 1.85% among those the imputed limit admits, and 0.26% among those the peak-window rule admits at essentially the same retention. That last figure is an extrapolation from scored cases to unscored records and is labelled as one; Section 4.13 tests the exchangeability it assumes.

At the level of a regional aggregate the choice of rule barely matters: admitting incomplete years at all raises the archive's mean annual peak by about half a degree, and the four rules differ by less than 0.1 °C (+0.52, +0.51, +0.53 and +0.45) and by 0.4 points in the share above 20 °C (+4.55 to +4.17).

The same is true of a multi-year slope. Theil–Sen slopes of the annual peak have RMSE 0.143 °C yr^{-1} on recorded days, 0.044 with reconstruction and no screening, 0.048 under length screening and 0.041 under the screen: the dominant step is again whether the record is repaired. Eleven complete years per station is in any case too short to resolve a decadal trend, so we report the slope error as a diagnostic of the reconstruction and make no trend claim.

Per station-year the choice does matter, and Section 5.5 follows it into the judgement an agency actually makes.

4.13 The conclusion survives instrument error, a stricter fitting protocol, and transfer across agencies

Missingness is weakly informative, and the effect is bounded well below the tolerance. Every case transplants a mask onto a year whose values were observed, which is valid only if the mask is ignorable in the sense of Rubin (1976); loggers are lost to floods, buried by ice and fouled in low flow, and each of those coincides with anomalous conditions. Across the 6472 real gaps, the mean anomaly of the seven days preceding a gap, relative to the station's own day-of-year climatology, is +0.018 °C for gaps of a week or less ($p = 0.44$), +0.038 for 8 to 29 days ($p = 0.56$), −0.002 for 30 to 89 days ($p = 0.98$) and −0.215 for 90 days and longer ($p = 0.081$). Donors recording through

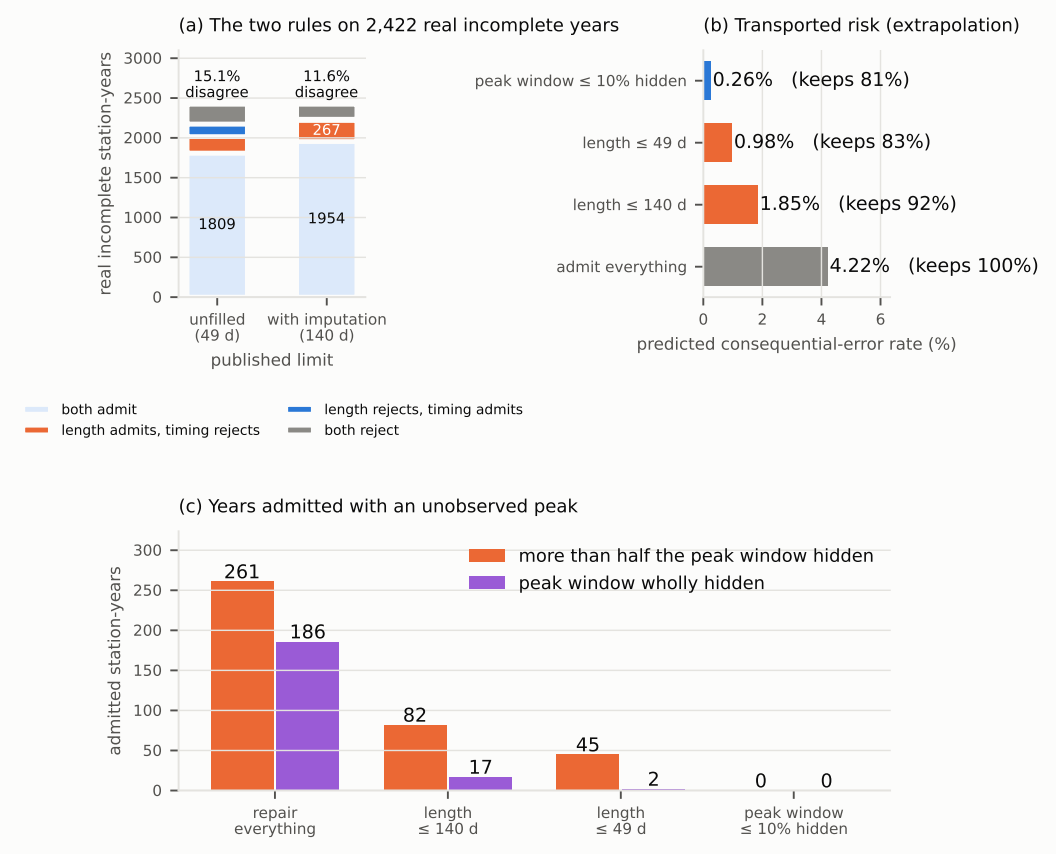


Figure 10. The rules on 2422 real incomplete station-years. (a) Agreement and disagreement at each published limit. (b) Transported risk of each admitted set. (c) Station-years admitted with the peak window more than half, or wholly, unobserved.

a gap run +0.063 °C warm during short gaps, detectable at this sample size ($p = 0.017$) and 3% of the within-station anomaly scale. The design is therefore defended on the magnitude of the effect rather than on its absence, and the class that warrants a limitation is gaps of three months and longer. A pattern-mixture check turns that defence into a bound: shifting every filled day by δ °C moves the completed year's peak by δ times the share of the realised peak window the gap hid, and across $\delta \in \{-1, -0.5, -0.25, 0, +0.25, +0.5, +1\}$ °C the screen keeps its advantage at equal retention with a network-clustered interval excluding zero in all three populations (development, the retrospective reserved panel, and the sealed USGS panel); no bias inside that grid erases it (`amendments/mnar_pattern_mixture/`). The assumption is bounded rather than verified, which is the strongest statement these records support.

Three further checks move the numbers and not the conclusion. Instrument error in these archives is systematic and time-varying as well as random (Jackson et al., 2025), which is why the check below is repeated under correlated, drifting and step-like error and not only under white noise. Perturbing the reference series by 0.15 °C of normal noise moves the development contrast from 5.98 to 6.03 points, the reserved contrast from 6.21 to 6.22 points, and the mean absolute error from 0.2361 to 0.2363 °C; the paired change is +0.049 points (−0.036, +0.150) on development and +0.013 (−0.175, +0.185) on the reserved panel, so the check no longer reports a reduction in the advantage (an earlier version evaluated the two arms with two different reconstruction choices). Correlated error displaces the 30-day endpoint harder at the same daily scale and still leaves the contrast within its interval: an AR(1) process with $\rho = 0.7$ by 2.25 times and a mid-year 0.3 °C step by 2.12 times, giving 5.95 and 6.21 points (development) and 6.19 and 6.43 (reserved). A constant annual bias of ±0.5 °C is an exact endpoint shift, so it moves the contrast by at most 1.0 point (largest change interval 0.35 to 1.82 points, on the reserved panel at −0.5 °C) while raising the mean absolute error to 0.30 °C. Restricting estimators to years before the gap raises reconstruction error from 0.242 to 0.270 °C on a panel that falls to 33633 development cases, and moves the contrast from −6.35 (−7.12, −5.52) to −5.92 points (−6.72, −4.99) with coverage 0.906. Under leave-one-provider-out the contrast is negative with an interval excluding zero for all nine agencies, from −11.6 points (Finland) to −1.2 (Switzerland), with conformal coverage between 0.82 and 0.96.

5 Discussion

5.1 What process explains the transition

The transition measured in Table 4 is not a property of rivers, of the estimators, or of the screening rule. It is a property of the functional being computed, and it follows from where that functional puts its weight. A statistic whose influence is concentrated on a few days is destroyed by losing those days and is almost unaffected by losing any others; a statistic whose influence is spread evenly over the year tends to degrade with how many days go missing and is less sensitive to which. Between those two limits the influence is neither concentrated nor uniform, and the discriminating power of a narrow conditioning window falls as the weight it captures falls. On this panel's exploratory width sweep, a fixed 30-day window stops beating a whole-year count near a three-month support; that numerical crossing is panel- and climate-dependent, not a universal constant.

Two measurements distinguish that account from a re-description of the data. First, the gradient ratios swap by roughly an order of magnitude in each direction between the peak and the annual mean, which is a direct measurement of how error grows along each axis rather than an inference from discrimination. Second, and more decisively, the account makes a prediction about a case it was not built on. The sine-fit phase has an influence function with two broad lobes a quarter-year from the peak, so the account predicts that peak-window overlap should be a poor axis for it — and it is, at 0.583 against 0.836 for gap length. A description fitted to the extreme-versus-average contrast would not have predicted that.

The same account explains where it failed. We predicted that conditioning on the estimand's own support would always be best, and it is not, because the support has to be estimated and the estimate wanders. Once the wander is measured the failure becomes a quantitative statement rather than an exception: the gain from widening the conditioning window tracks the wander from 16.7 days at $k = 1$ to 7.4 days at $k = 90$ and disappears when the support becomes wide enough to contain it. This is the part of the result we would expect to transfer to other variables and other climates, because nothing in it is specific to water temperature: any estimator conditioned on an estimated support inherits the uncertainty of that support.

5.2 Why this agrees with the standing guidance rather than against it

An earlier framing of this work held that the incumbent rule gets an ordering backwards. Section 4.6 shows that reading is wrong on the guidance's own terms. Johnson et al. (2021) derive their limits for sine-fit mean, phase and amplitude, and for those three estimands a completeness threshold is the correct axis or within 0.02 AUC of it. What we correct is not the threshold but the habit of carrying it across to an estimand it was not derived for.

That reading also resolves the anomaly the guidance itself reported. Its authors observed that the timing of consecutive missing data had "inconsistent effects ... among regions, years, and individual parameters" on annual thermal-signal estimates. Their three estimands have three different influence functions — a constant, two lobes at the extremes, two lobes at the transitions — and only one of the three has weight concentrated narrowly enough for a timing effect to appear. Observing all three at once, inconsistency is the only thing one could see. We do not contradict the observation; we decompose it, and we can now say in advance which of their parameters should show a timing effect and which should not.

A smaller result points the same way and was not expected. The fixed information-conditioned policy gets slightly *worse* at donor-less stations once air forcing becomes available, because its preference order was written when air was rare and it now prefers an air regression to a climatology at sites where the climatology is better. A fixed ladder over estimators is not robust to a change in what is available; a per-case selection is. That is an argument for the method of this paper rather than against it, and we report it because it was the opposite of what we expected.

The disagreement with the gap-filling literature is of a different kind and is not resolved by our result. Beaufort et al. (2022) fill every record alike, and on our panel pairing the incumbent rule with that treatment makes it worse rather than better (Table 9). That is not evidence against air-temperature regression, which is the best available estimator at many of our stations and which the screen selects for 20.0% of cases. It is evidence that a reconstruction chosen once for a whole archive and a reconstruction chosen per station-year are different objects, and that the quantity a screening comparison must hold fixed is the second.

5.3 What generalises, and what does not

Three parts of the result are portable. The governing quantity of Equation (3) is defined wherever a derived statistic is computed from an incomplete record, and its ingredients — an influence function and a mask — are available in any such setting. The protocol that measures the transition is portable in the same way: compute the statistic over a family of support widths, measure how far the support's location moves between periods, dilate by that amount, and condition on the result. And the finding that a screening rule must be specified together with its strata is a statement about selection, not about temperature.

Three parts are not. The numerical crossover near three months depends on how concentrated a river's thermal year is and on how far its peak wanders between years; both are properties of this climate and this variable. The operational gain is regime-dependent: the sealed USGS confirmation recovers a pooled contrast of -7.9 points but the empirical arm only -1.7, and the multi-agency transport panel, while negative as a whole (-4.6, $[-6.5, -3.3]$), carries an empirical arm of -1.9 ($[-3.3, -1.0]$) — real, but far smaller than the designed-outage figure an agency might quote. Catchment associations rest on 140 of 204 stations in one country and should be read as attribute correlations on that subset, not as a geology-versus-terrain law.

One boundary deserves naming separately, because it limits interpretation rather than transfer. The mechanism argument is supported by an influence-function derivation, by measured error gradients, by a continuum with a located crossover and by an out-of-sample prediction about the phase. It is not supported by an independent observation of the internal quantity itself, because the hidden mass of Equation (3) is by construction unobservable on the records it matters for. A different representation that reproduced the same gradients would not be excluded by our evidence.

5.4 What the remaining uncertainty changes, and what it does not

Uncertainty enters this study in four places, and it is worth separating what each one moves. Measurement error in the reference series changes the numbers in the third decimal and changes neither the mechanism nor the recommendation: $0.15\,^{\circ}\mathrm{C}$ of noise moves the reserved contrast from 6.21 to 6.22 points, and time-correlated error displaces the endpoint twice as hard while still leaving the contrast inside its interval. Fitting pro-

tocol changes the numbers by about half a point and changes nothing else: restricting estimators to prior years moves the contrast from −6.35 to −5.92. Neither of these is capable of overturning a conclusion.

The other two are. The choice of population changes the practical conclusion completely: on the arm built from the gap geometry these networks actually experience the effect is −0.71 points with an interval containing zero, and any reader who needs a number for their own archive needs the arm that matches their gap regime rather than the pooled figure. And the absence of an independent observation of the internal quantity limits the mechanism judgement, as the previous section states: we can show that conditioning on an estimate of the support discriminates better, and we cannot show that no other representation would do as well.

Two further limits belong here because they are specific rather than generic. Missingness is weakly informative and the transplant design is defended on the magnitude of that effect rather than on its absence, which makes the defence weakest for gaps of 90 days and longer, where the leading flank runs 0.215 °C cold. And the archive-level consequence is small: the choice of rule moves a regional mean annual peak by less than 0.1 °C. We report that because it deflates a claim we might have made. At the level of a regional aggregate the decision that matters is whether to repair; the rule matters per station-year, and it matters for who is in the archive.

5.5 What changes in the judgement an agency actually makes

The archive-level aggregate is the wrong place to look for the value of a screening rule, because it averages over the station-years where the call is easy. A criterion-exceedance judgement is the right place, because it is made one station-year at a time and it is where a wrong peak becomes a wrong decision. On reserved networks the misclassification rate of a 20 °C exceedance is 6.86% when the statistic is computed on recorded days, 1.97% when every year is reconstructed and not screened, 1.53% under length screening and 0.36% under the screen. Restricted to the years where the call is close the four figures are 16.84%, 9.50%, 7.26% and 1.92% within 1.0 °C of the threshold and 19.10%, 13.12%, 10.24% and 3.38% within 0.5 °C, so the advantage is proportionally larger exactly where it is needed. The bias in the count of years above 20 °C falls from −0.76 years per station under recorded days and −0.32 under length screening to +0.06.

That result is not an artefact of the threshold chosen. Swept from 14 to 24 °C the screen's misclassification stays between 0.18% and 0.37% while the length rule stays between 0.84% and 1.53%. The sweep matters because no single number is the criterion: regulatory practice uses a seven-day average of daily maxima with criteria differing by several degrees between jurisdictions and designated uses, 16 °C for core cold-water salmonid habitat and 13 °C for salmonid spawning (Washington State Department of Ecology, 2024), which is why the seven-day endpoint is reported throughout and why the threshold is swept rather than asserted. The illustration is not a compliance test, and we can now say by how much it misses one: on the 1591 station-years of the sealed USGS panel that publish both statistics, the regulatory seven-day average of daily maxima sits a median +1.57 °C above our seven-day average of daily means (5th to 95th percentile +0.49 to +4.43), and at criteria of 16, 18 and 20 °C the two endpoints disagree on 5.5%, 4.3% and 10.4% of station-years — every disagreement a missed exceedance and none a false one (`amendments/regulatory_daily_max/`). A threshold shift matches counts but not decisions, because the offset is nearly constant (slope −0.002 per °C). An agency applying a daily-maximum criterion must compute it from daily maxima, which the archive publishes for 90 of the 91 sealed stations.

For a judgement of this kind, screening is not the best use of the calibrated object. Discarding a station-year is a blunt response to uncertainty about a single call: screening at the 0.5 °C tolerance judges 67.5% of years at 0.36% misclassification, whereas abstaining only where the interval straddles the threshold judges 92.7% at 0.42%, and doing both judges 65.7% at 0.08%. Keeping 93% of station-years rather than 68% at essentially the same error rate matters to anyone assembling a regional archive. The two actions answer different questions — screening decides what enters an archive, abstention decides what one judgement may say — and the same calibrated object serves both, which is an argument for calibrating rather than for screening.

5.6 What an agency should change, and in what order

The first recommendation is to treat screening as estimand-specific. An agency assembling warm-season extremes from maximum gap length should trial conditioning on how much of a station's climatological warm window the gap hides, and should keep length-based rules for annual means and the sine-fit regime parameters for which standing guidance was derived. Cold extremes are not an exception to this: the support axis has to

be defined on the cold season, and when it is, it beats the length rule for the coldest 30-day mean on the development panel; a warm-season support axis is the wrong tool there, which is what the reversal in Table 3 shows. On the retrospective reserved holdout, substituting a support-aware screen takes the consequential-error rate from 9.7% to 2.5% at equal retention on pooled designed outages; the empirical-arm gain is not resolved, so this is not yet a claim about the networks' natural failure calendar.

Maintenance-calendar pricing that previously appeared to show autumn servicing gains of several percentage points is withdrawn from recommendations until the cross-year outage placement is recomputed: the earlier counterfactual truncated late-year gaps at day 366 while still labelling them by their nominal length. Until that recompute is confirmed on networks outside the development programme, schedule changes remain a hypothesis suggested by autumn-heavy long outages, not a verified error-rate benefit. It cannot be replaced by an observed counterfactual on this archive, and we checked rather than assumed that: for the 200 USGS stations of the incomplete-year population we classified all 2921 real interior gaps and 68180 missing days against the provider's own daily-value qualifiers, and no cause-naming code — maintenance, equipment, ice, seasonal or discontinued — occurs once. What the qualifiers do support is an archival decomposition: 94.9% of missing days are days the provider published no record for at all (interval 88.7 to 98.2), 4.8% were published only under a different statistic of the same parameter, and 0.3% were published for the target statistic and dropped by the archive (`amendments/outage_causes/`). A rule that excludes planned-maintenance gaps and keeps equipment failures is therefore a no-op on 0 of 1361 station-years, while re-ingestion would recover five station-years (all 2024–2025) whose target values the live service now serves. Redundancy investments similarly need cost and failure-mode evidence beyond synchrony correlations on the opened panel.

Thresholds should be applied within strata when difficulty correlates with elevation or donor poverty, or the archive can quietly lose mountain stations. Nothing here applies to ungauged sites, because locating support uses a station's own history. A pooled conformal interval is a statement about a population of networks, not an automatic guarantee at a single new site; cluster-aware calibration must be chosen on calibration data only and validated on networks outside the development programme before it is sold as site-level coverage.

6 Conclusions

Whether a data gap matters depends on how much of the target statistic's support it hides. That quantity is a useful risk descriptor and screening axis for archive decisions; we do not claim an unconditional proportion law relating hidden support to loss for every residual structure and every competing peak window. A frozen screen, scored as a retrospective network holdout on 36 held-out networks, reduces consequential error for the hottest-30 mean from 9.7% to 2.5% at 70% retention on pooled designed outages. Continuum, crossover and dilation results that suggest when timing versus length should govern were developed after that panel was opened and remain exploratory until confirmed on networks outside that programme.

Three conditions bound what can be said now. On the sine-fit estimands standing guidance was derived for, transferring a warm-season threshold is the mistake to avoid rather than the guidance itself. Operational gain under the agencies' own outage geometry is not established on the empirical arm. Difficulty-based screening can thin high, steep, donor-poor catchments unless thresholds are applied within strata. Maintenance-calendar benefits and "annual peak never observed" counts are not retained as verified effects: the former require a corrected multi-year outage geometry, and the latter currently flag wholly masked *climatological* windows.

What follows for practice is correspondingly limited. Prefer estimand-specific rules; keep length-based completeness for averages and sine-fit regime parameters while trialling support-aware rules for warm-season extremes; report retention by catchment attribute alongside error rates; deploy only prediction interfaces that do not require true errors at decision time; and treat autumn maintenance and regulatory hottest-7 claims as unvalidated until recomputed on the correct endpoints and geometries.

Open Research Statement

Daily water-temperature records are from nine public providers (ARSO, CHMI, FOEN, GKD Bayern, LUBW, NVE, Rijkswaterstaat, SYKE and the USGS), plus SMHI and the UK Environment Agency for the transport panel; none required a data-use agreement. Nine of the eleven publish under a named open licence (US public domain; CC BY 4.0 for CHMI, GKD Bayern and SYKE; OGL v3.0 for the UK EA; opendata.swiss terms for FOEN; NLOD/CC BY for NVE; CC BY 4.0 SE for SMHI; Licence Ouverte

for Hub'Eau). Two state terms without naming a licence (ARSO requires attribution on publication; Rijkswaterstaat requires source and retrieval date). One, LUBW, reserves copyright: its raw values are therefore *not* redistributed with this paper, and the seven LUBW networks are available only through LUBW's own service; the analysis reads them locally and reports derived statistics. Licences, acquisition routes and verbatim evidence are in `metadata/data_rights.csv`, `results/wrr_submission_ready/measurement_provenance_dictionary.csv` and `current_data_acquisition.csv`. Station coordinates are from those providers' own catalogues, the USGS NWIS site service, and ARSO's public station feed. Daily 2 m air temperature is NASA POWER (MERRA-2) (NASA Langley Research Center, 2026). Basin characteristics are GAGES-II (Falcone, 2011).

All analysis code, derived artifacts, figures, the editable draw.io sources of the schematic figures and the frozen pre-registration are in the accompanying repository. A claim registry (`results/wrr_submission_ready/claim_registry.csv`) records each headline claim's artifact, endpoint, unit, denominator, evidence type and whether it may enter the abstract or conclusions. An automated audit checks a declared set of archived manuscript numbers against artifacts; it is not yet a full semantic binding of every live TEX numeral. Input-file hashes for the 146 study networks are in `results/wrr_submission_ready/current_data_manifest.csv`. A permanent archive of the repository and a DOI will be deposited on acceptance.

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